

Software Intern Assignment

Ground Station Telemetry Dashboard

You are provided with telemetry data collected from underwater robotic vehicle along with timestamps from an onboard camera system.

Your task is to process the telemetry data, synchronize it with camera data, and build a visualization dashboard similar to a Ground Control Station (GCS).

Dataset Link: [dataset link](#)

Dataset Provided:

1. telemetry.csv
 - Vehicle telemetry data
2. image_timestamps.csv
 - Mapping of image filenames to timestamps
3. frames/
 - Folder containing 10 camera images
 - Image names: frame_001.jpg to frame_010.jpg

Task 1: Telemetry Processing

Using the provided telemetry dataset, calculate:

- Total Distance Travelled
- Average Speed
- Maximum Speed
- Mission Duration
- Packet Loss Statistics (if detected)
- Communication Health Summary

The application should handle missing or inconsistent telemetry records gracefully.

Expected Output

Example:

Mission Summary

Mission Duration: 00:12:35
Total Distance Travelled: 2.34 km
Average Speed: 3.12 m/s
Maximum Speed: 5.87 m/s

Communication Summary
Packets Received: 115
Packets Missing: 5
Packet Loss: 4.17%
Communication Status: GOOD

Task 2: Camera-Telemetry Synchronization

Using image timestamps and telemetry timestamps:

- Match each image with the nearest telemetry record.
- Associate image metadata with telemetry information.

Expected Output

For each image:

Image	Timestamp	Latitude	Longitude	Depth	Speed	Battery
frame_001.jpg	10:00:05	28.xxxx	77.xxxx	15.2 m	3.2 m/s	95%

The candidate may:

- Display this as a table
- Show within the dashboard

Task 3: Ground Station Dashboard

Build a dashboard to visualize mission telemetry data. The dashboard should:

Display the vehicle trajectory using latitude and longitude coordinates.

Show the current vehicle status, including:

- Latitude
- Longitude
- Speed
- Battery Level
- Signal Strength
- Timestamp

Visualize the mission path with:

- Start Position

- Current Position
- End Position

Allow users to select any point on the trajectory and view:

- Associated telemetry information
- Nearest synchronized camera image

Display the synchronized camera image along with its corresponding telemetry details.

Bonus (Optional)

Interactive map visualization (Leaflet, Mapbox, OpenLayers, Cesium, etc.)

3D trajectory visualization using depth data

Time-range filtering and mission playback

GitHub Repository

Upload complete source code to a GitHub repository.

Deliverables

1. GitHub Repository Link
2. Demo Video Link
3. README Documentation